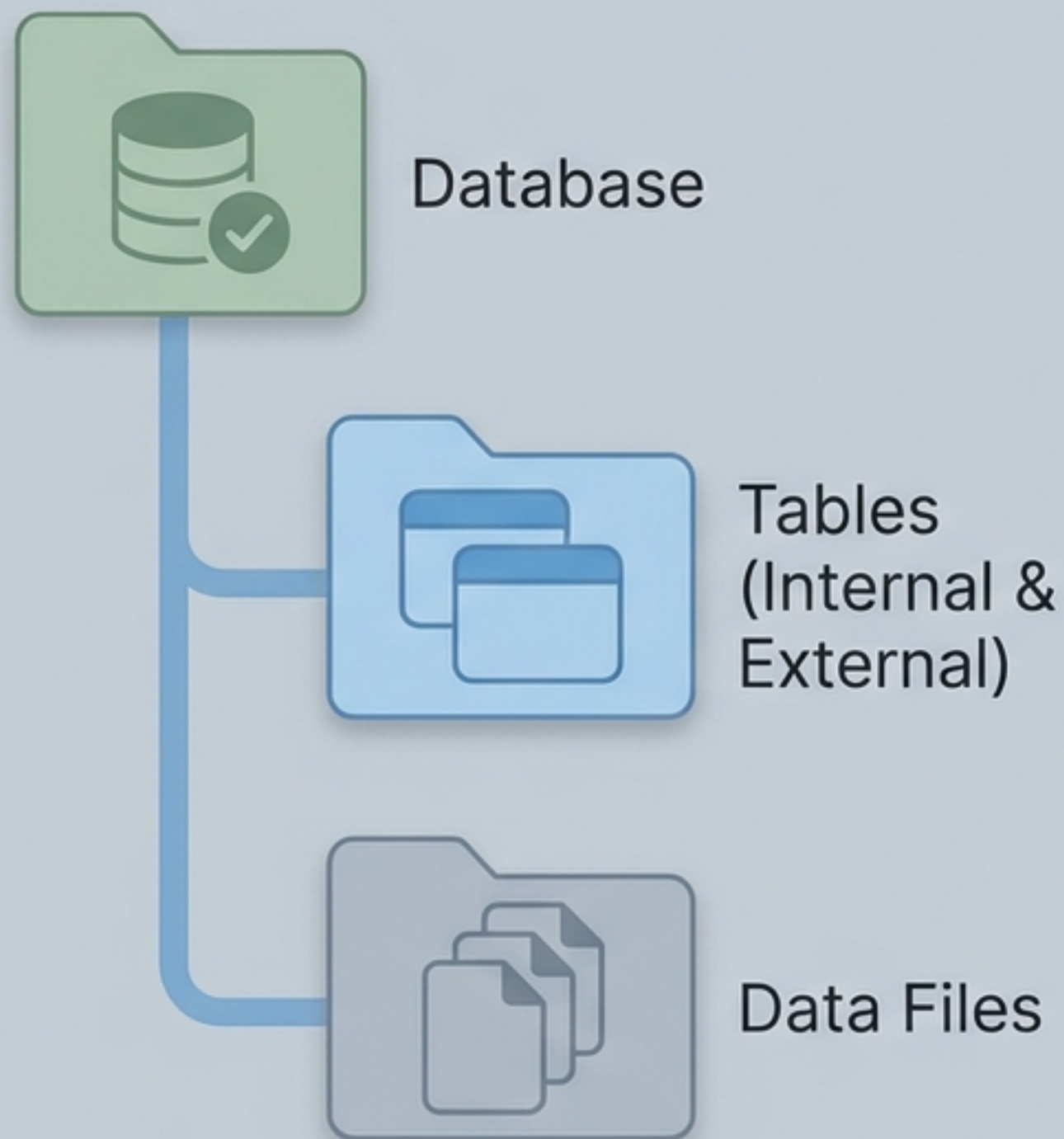


The Apache Hive Practitioner's Playbook

Complete Study Notes for Architecture, Ingestion, and Advanced Analysis



The Hive Storage Architecture



```
CREATE DATABASE db_name;
```

```
SHOW DATABASES;
```

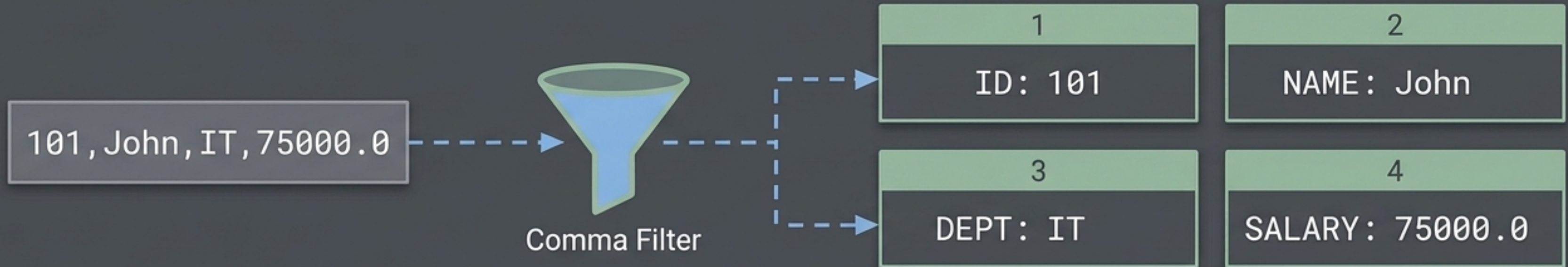
```
USE db_name;
```

```
DROP DATABASE db_name;
```

Crucial Step: All operations require specifying an active database using the USE command before manipulating any tables.

Defining Managed Tables & The Parsing Engine

The Parsing Engine Graphic



```
CREATE TABLE employees (  
  id INT,  
  name STRING,  
  department STRING,  
  salary DOUBLE  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ',';
```

Unlike MySQL's VARCHAR, Hive uses STRING

Data is read line-by-line. The FIELDS TERMINATED BY clause tells Hive how to split a single string into individual data columns.

Architecture Matrix: Internal vs. External Tables

Internal (Managed) Tables	External Tables
Keyword Default creation (No extra keyword required)	Keyword CREATE EXTERNAL TABLE
Storage Generates a complete data backup in HDFS	Storage No backup generated; acts as a temporary data pointer
Location /user/hive/warehouse/db_name.db/table_name/	Location Custom HDFS path defined by LOCATION '/hdfs_directory/'
Use Case Permanent storage managed entirely by Hive architecture	Use Case Temporary access to external data sets without moving source files

Data Ingestion Paths

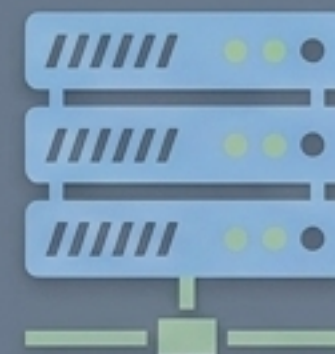
Local Mode



```
1 | LOAD DATA LOCAL INPATH  
2 |   '/home/user/file.txt'  
3 | INTO TABLE employees;
```

Uses the LOCAL keyword because the file originates from the local Linux file system.

HDFS Mode



```
1 | LOAD DATA INPATH  
2 |   '/test1/file.txt'  
3 | INTO TABLE employees;
```

Omits the LOCAL keyword. The path points directly to an existing HDFS directory.

Core Analysis & Aggregation

Selection

View all table data

```
SELECT * FROM employees;
```

Uniqueness

Find unique records

```
SELECT DISTINCT department...
```

Filtering

Apply logic conditions

```
WHERE salary > 50000  
AND department = 'IT'
```

Sorting

Ascending or descending order

```
ORDER BY salary DESC
```

Grouping

Aggregate per category

```
GROUP BY department
```

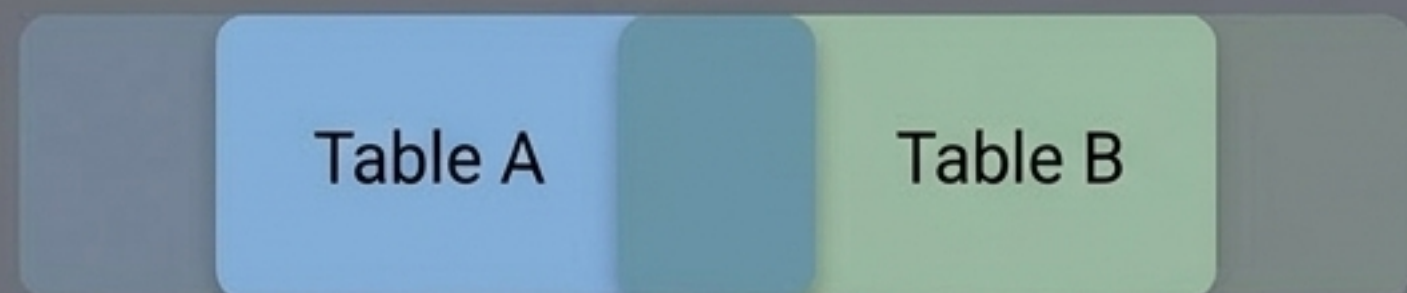
Post-Aggregation

Filter applied after grouping

```
HAVING avg(salary) > 50000
```


Relational Merging: The Core Joins

Inner Join



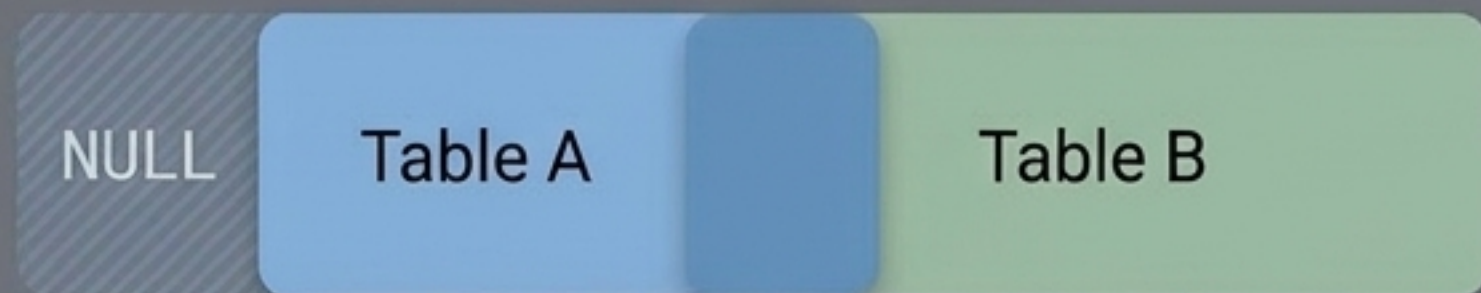
Only matching records remain.

Left Join



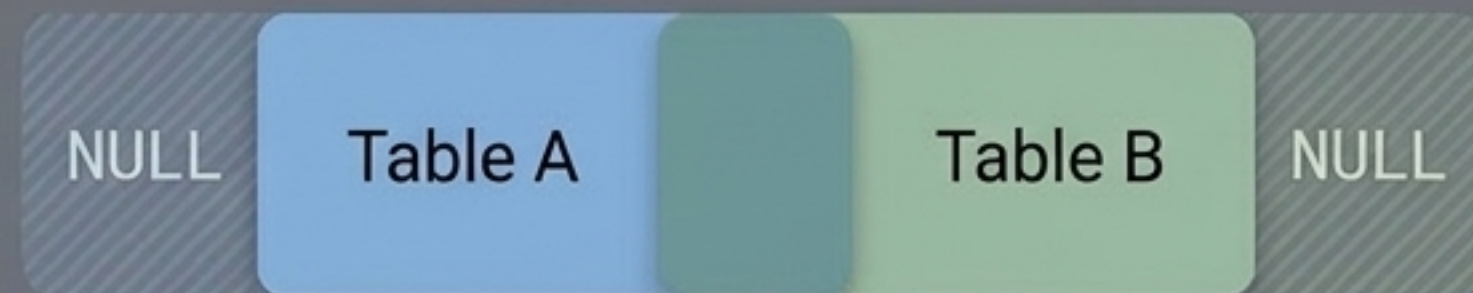
All left records kept. Unmatched right is NULL.

Right Join



All right records kept. Unmatched left is NULL.

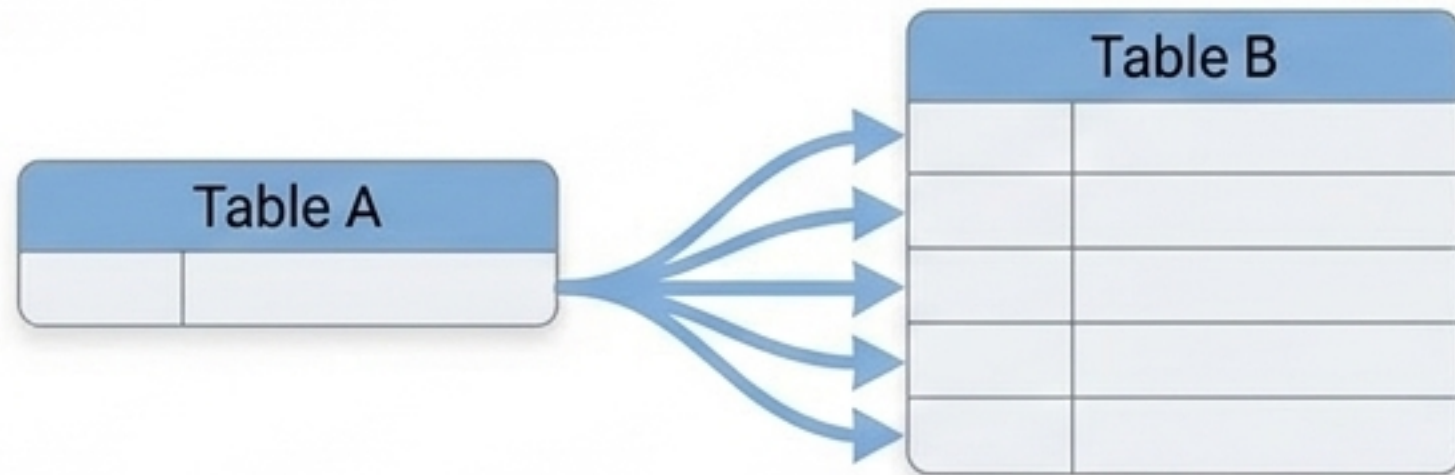
Full Outer Join



All records kept from both sides.

Advanced Relational Logic: Cross & Self Joins

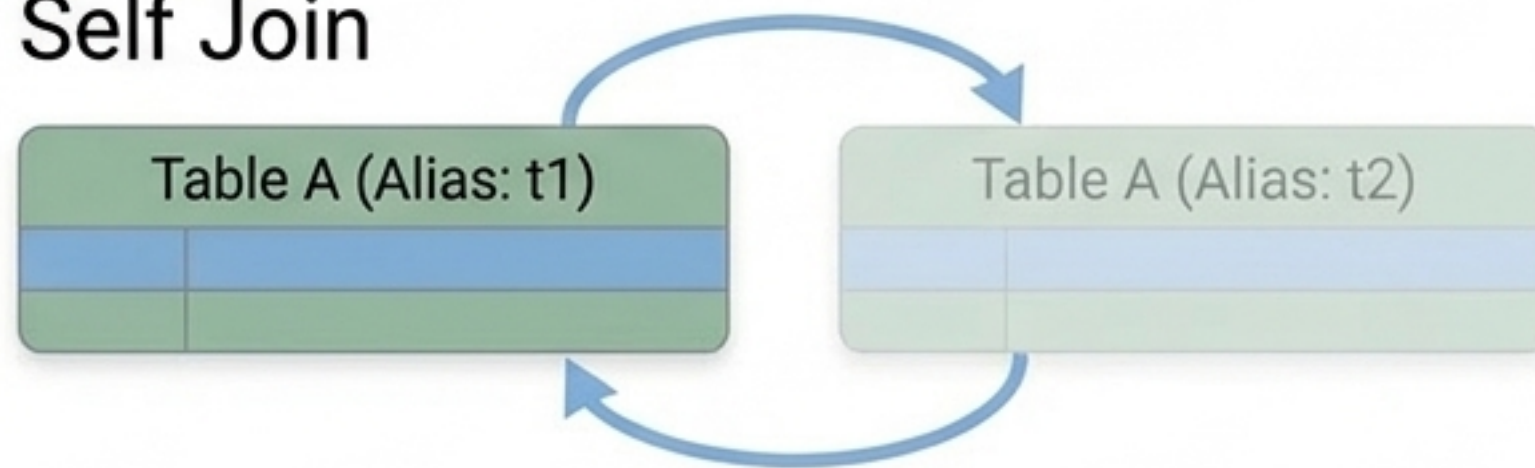
Cross Join (Cartesian Product)



No Analytical Benefit.

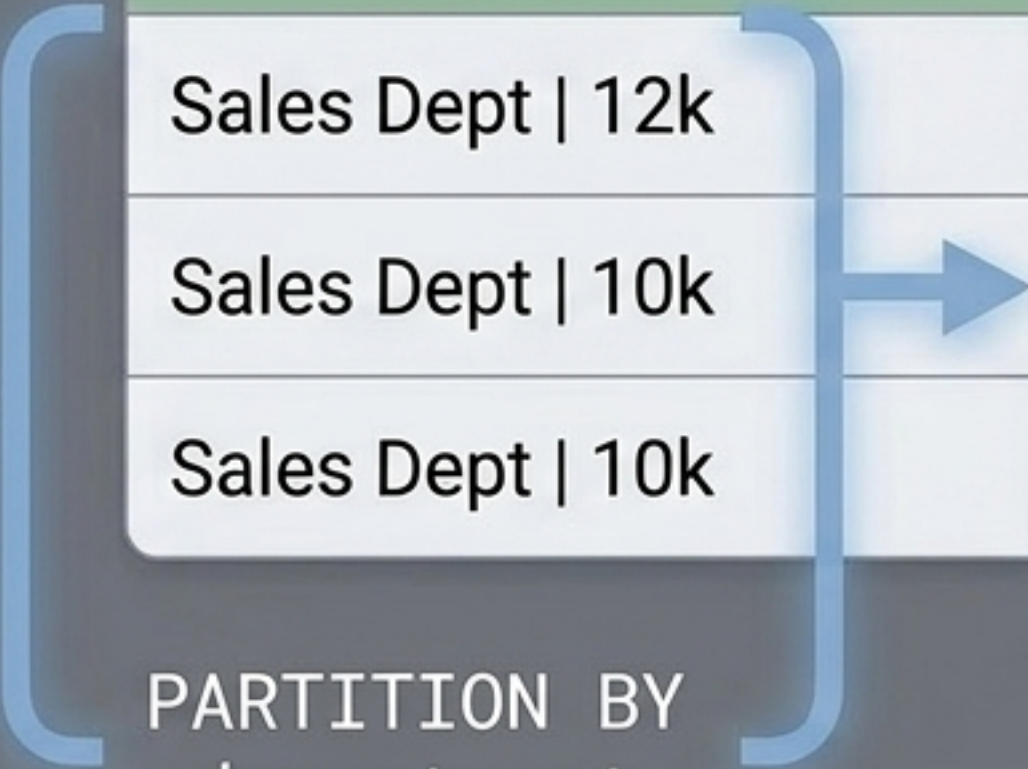
No Analytical Benefit. The weament. Evaluates every possible combination. Primarily a part of syntax rather than a practical analytical tool.

Self Join



Joining a table to itself using distinct aliases. Used to compare rows within the same exact table.

Anatomy of a Window Function



	SUM(salary) OVER(Window)
Sales Dept 12k	32000
Sales Dept 10k	32000
Sales Dept 10k	32000

PARTITION BY
department

Unlike standard **GROUP BY** clauses which collapse rows into a single output,

Window Functions perform calculations across a partitioned set of related rows while preserving the original row display.

The Window Functions Duel

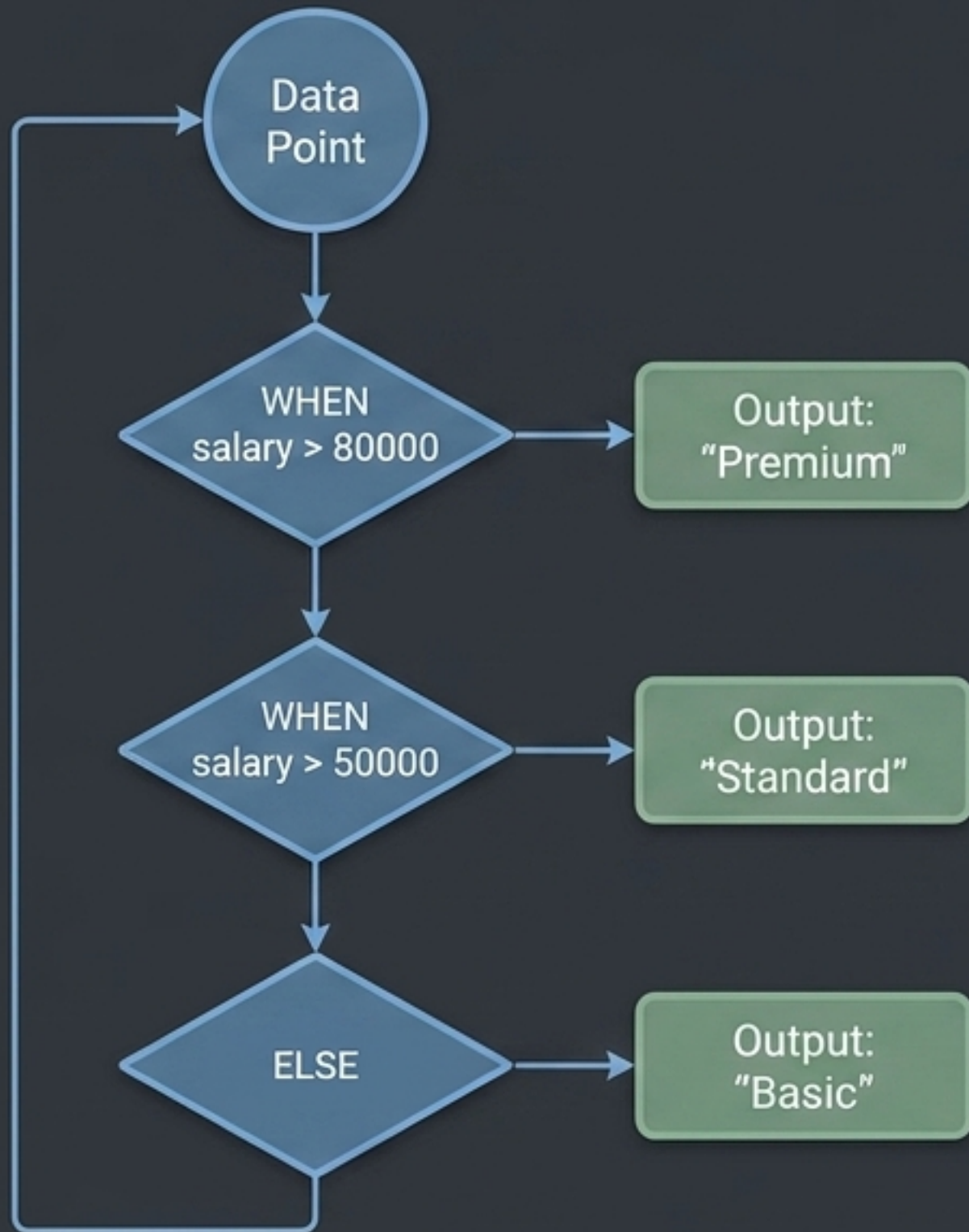
Base Data	ROW_NUMBER()	RANK()	DENSE_RANK()
Value A	1	1	1
Value A	2	1	1
Value A	3	1	1
Value B	4	4	2

Ignores duplicates,
maintains pure
sequential list.

Identifies ties, skips
subsequent sequence
numbers.

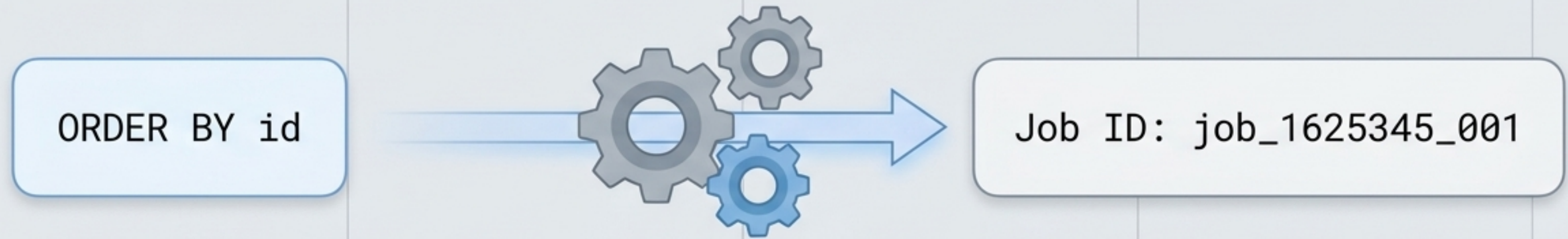
Identifies ties, maintains
consecutive numbering
without skipping.

Conditional Logic: Case Statements



```
SELECT name, salary,  
CASE  
  WHEN salary > 80000 THEN 'Premium'  
  WHEN salary > 50000 THEN 'Standard'  
  ELSE 'Basic'  
END AS salary_tier  
FROM employees;
```


Under the Hood: The MapReduce Reality



High-level Hive SQL queries do not execute instantly. They are translated into MapReduce jobs that run asynchronously on the Hadoop cluster.

Symptom	Action	Resolution
Job ID is stuck (e.g., executing for 2 hours instead of 10 minutes).	Copy the assigned Job ID and share it directly with the cluster administration team.	Admins will audit cluster health, terminate obstructing processes, or kill the stuck job to allow a manual re-run.